SCIENTIFIC ARTICLE

Metacarpal Fractures in the National Football League: Injury Characteristics, Management, and Return to Play

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Purpose This study investigated metacarpal fracture occurrences, characteristics, treatments, and return-to-play times for National Football League (NFL) athletes.

Methods NFL players who sustained metacarpal fractures during the 2012 to 2018 seasons were reviewed. All players on the 32 NFL team active rosters with metacarpal fractures recorded through the NFL Injury Database were included. Player age, time in the league, player position, injury setting, injury mechanism, fractured ray, management, and return-to-play were recorded.

Results There were 208 injury occurrences resulting in 1 or more metacarpal fractures, identified in 205 players. Of these, 81 (39%) injuries were operated. Return-to-play data were available for 173 (83%) injured players. The median return-to-play time for all athletes was 15 days (interquartile range, 1–55 days). Of the injured players, 130 (71%) missed time but returned the same season. Within this 130-player subset, 69 (53%) were treated nonsurgically and 61 (47%) operatively with median return-to-play times of 16 days (interquartile range, 6–30 days) and 20 days (interquartile range, 16–42 days) respectively. Eighteen individuals in this 130-player subgroup sustained a thumb metacarpal fracture. The return-to-play time was slower for patients sustaining thumb metacarpal fractures compared to other metacarpal fractures, and was significantly longer (median, 55 days) following nonsurgical treatment of thumb fractures compared with operative intervention (median, 24 days). A regression analysis revealed no trend or difference in return to football with respect to player age, time in the league, injury setting (practice vs game), injury mechanism, articular involvement, multiple concomitant injuries, or player position.

Conclusions Most NFL players who sustain metacarpal fractures miss less than 3 weeks and return to play the same season. The only variables that lessen the return-to-play time are involvement of lesser digit metacarpals and operative intervention for treatment of thumb metacarpal fractures. *(J Hand Surg Am. 2022;■(■):1.e1-e8. Copyright © 2022 by the American Society for Surgery of the Hand. All rights reserved.)*

Type of study/level of evidence Prognostic IV.

Key words Mechanism of injury, metacarpal fracture, NFL, return to play.

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Metacarpal fractures frequently happen during contact sport participation and account for substantial player morbidity and lost time from sports.[1–6] In football, the time lost may vary based on position and ball-handling requirements. For example, a lineman can more easily function with a club cast or other means of external support, whereas a receiver, who requires an unencumbered hand, may benefit from internal stabilization using a plate-and-screw construct. In either case, a rapid return to play is typically a goal shared by the team and the player.

Two prior studies reported on metacarpal fractures in a National Football League (NFL) population.[7,8] Ellsasser and Stein[7] published their experience treating players from 1 NFL team from 1962 to 1976. Without access to a formal database, they found only 8 metacarpal fractures in 7 players. Return to play for the 2 operated players was within a week of surgery, and the authors recommended "early and aggressive surgery for intra-articular and certain (displaced) metacarpal fractures."[7] Mall et al[8] performed an epidemiologic investigation into hand, first ray, and finger injuries, as 3 separate injury cohorts, in NFL players between 1996 and 2005. They used a now-replaced injury surveillance registry that defined injury occurrence strictly by the player terminating or missing at least 1 training session, practice, or game event recorded by NFL training staff (NFL Sports Injury Monitoring System, Flantech Group). The data allowed the authors to register the injuries within the categories mentioned but did not provide information on a player-specific or injury-episode basis.

A more comprehensive Injury Surveillance System (ISS) was mandated prior to the 2012 and 2013 season and combined clinical management with the adoption of the current electronic medical record (EMR) system before the 2015 and 2016 season. The NFL's EMR records all injuries sustained by players, even in cases where neither a practice nor a game is missed.[9] For example, the EMR has revealed that an average of 637 hand injuries occur every year in the NFL, despite the fact that many do not translate to missed time. Prior to this investigation, no study has evaluated the ISS, the detailed EMR, and the linked league-specific data to provide a precise analysis of individual NFL players sustaining metacarpal fractures.

The goals of this study were to use the detailed league-wide data to specifically describe metacarpal fracture occurrences and mechanisms of injury, as well as player- and injury-specific variables. The secondary aim of this investigation was to identify descriptive elements that have an impact on return to play and, when considered with other external factors, may provide guidance to team physicians as they seek to recommend the safest and most efficacious treatment for their patients.

MATERIALS AND METHODS

This was a retrospective, observational investigation conducted by the NFL Musculoskeletal Committee's hand and wrist subcommittee, submitted through the Medical Research Application Process and approved by the NFL Players Association, as well as the Institutional Review Board of the Icahn School of Medicine at Mount Sinai. Study subjects included all players on the 32 NFL active team rosters who sustained metacarpal fractures during the 2012 through 2018 NFL seasons, recorded through the NFL Injury Database, either via the ISS (2012–2014) or the EMR (2015–2018), then made available through IQVIA (Trademark IQVIA Holdings Inc). Included metacarpal fractures were those sustained at any point during practice or game play within the preseason, regular season, or postseason. For each team, the number of participating athletes included 90 in the first 3 preseason games, 75 in the fourth and final game of the preseason, and 53 in the regular season. When summed for all 32 teams, 2,760 players competed in the preseason, 1,696 in the regular season, and 1,908 in both the regular season and preseason per year. These numbers were consistent for all 7 years of the study.

Injured players were described by age, time in the league, player position, player activity, and setting (game, practice, or conditioning) at the time of injury. Injury characteristics included the ray involved, mechanism, articular involvement, and coexistent ipsilateral hand or wrist injuries. Operative or nonsurgical treatment was specified. The method of surgical stabilization and specific surgical considerations, such as fracture pattern, were not recorded. When available, return to play was determined based on days from injury to participation in practice or games. Hand dominance, radiographs, operative reports, and data pertaining to the duration of protective casts or orthoses after return to football were not recorded.

A statistical analysis was performed for means, medians, interquartile ranges (IQRs), and standard deviations. A t-test was performed to compare the times to return to play for those patients who underwent operative management and those who were treated without surgery. A chi-square test was

completed to compare the proportions of injuries that underwent operative intervention with respect to the metacarpal involved, articular involvement, player position, mechanism of injury, and timing of the injury within the season. A separate regression analysis was carried out to determine the impact of various independent factors on return to play. This was performed by using "return to play" in days as the dependent variable. The potential predictor variables were entered into the model in the following order: player age, years in the league, mechanism of injury, articular involvement, presence of concomitant hand and wrist injuries, player position, treatment (surgery vs no surgery), and metacarpal bone involved.

RESULTS

There were 208 injury occurrences recorded resulting in 1 or more metacarpal fractures, identified in 205 players over this 7-season study, with a consistent average of approximately 30 injury episodes per year (range, 24–37; Table 1). Three players sustained injuries on 2 separate occasions within this 7-year time frame. There were no refractures. The overall metacarpal fracture incidence per year was 1.5%, with calculated incidences of 0.4% in the preseason and 1.1% in the regular season (Table 2). There was a wide distribution of player activity at the time of injury, with most injury episodes occurring during participation on special teams (n = 38) or as a defensive back (n = 38). Of the 208 injuries, 162 (79%) were caused by player-to-player contact and 46 (21%) occurred due to contact made with the ground (Table 3). Of the 208 separate injury episodes, 197 (95%) resulted in isolated metacarpal fractures without concomitant ipsilateral hand or wrist pathology reported (Table 4). There was no correlation between player position and the location of metacarpal fracture or complexity of injury.

Overall, 81 (39%) players were managed operatively (Table 4). The presence of a thumb metacarpal fracture was predictive of having surgery: of the 26 thumb metacarpal fractures, 18 (69%) underwent operative fixation ($P < .05$), 2 of which were intra-articular (1 head, 1 base). Articular involvement, player position, mechanism of injury, and timing within the season were not predictive of having surgery. Though missed time was documented for 200 (96%) injury episodes, the actual number of days before return to play was recorded in only 173 (83%) cases (Tables 5 and 6). Of these 173 cases, 43 were recorded as not missing time because of their injury and 130 (75%) missed time but were able to return in the same season. The median number of days lost from competition after injury in these 173 players was 15 (IQR, 1–55 days). The median time to return to play for the 130-player subset was 18 days (IQR, 10–39 days). Of these 130 players, 61 (47%) underwent operative stabilization and 69 were managed nonsurgically. The median of 16 days to return to play (IQR, 6–30 days) for the 69 patients managed nonsurgically was less than the 20 days (IQR, 16–42 days) required for the 61 operated patients, but this difference was not statistically significant ($P = .23$).

A regression analysis was performed on the 130-player subset to determine which factors contributed to players missing time. Of these 130 players, 18 (14%) sustained thumb metacarpal fractures and 112 had nonthumb metacarpal fractures. The overall median time to return to play for the 18 players with thumb metacarpal fractures was 29 days (IQR, 16–58 days), compared to 17 days (IQR, 8–30 days) for players with nonthumb metacarpal fractures ($P < .05$). Of the 18 thumb metacarpal fracture

TABLE 1. Age at Injury, Injury Breakdown Per Season, Injury Setting, and Injury Time Within the Season for All 208 Injuries

Characteristic	n (%)
Age, mean (range)	26 (21–38)
Total injuries*	208
Per season	
2012–2013	25 (12%)
2013–2014	24 (12%)
2014–2015	37 (18%)
2015–2016	34 (16%)
2016–2017	32 (15%)
2017–2018	31 (15%)
2018–2019	25 (12%)
Injury setting	
Games	146 (71%)
Practice	60 (28%)
Unknown	2 (1%)
Injury time	
Preseason	78 (38%)
Regular season	126 (60%)
First 8 games	66 (32%)
Second 8 games	60 (28%)
Postseason	4 (2%)

*Three players sustained injuries on 2 separate occasions within the 7-year time frame (no refractures).

TABLE 2. Incidences of Metacarpal Fractures Per Season Comparing Preseason and Regular Season

Season	Total Injuries, n = 208 (Preseason, n = 78; Regular Season, n = 126; Postseason, n = 4)	Overall Incidence (1.5%)	Preseason Incidence (0.40%)	Regular Season Incidence (1.1%)
2012–2013	25 (10, 14, 1)	1.3%	0.36%	0.93%
2013–2014	24 (12, 12, 0)	1.3%	0.35%	0.90%
2014–2015	37 (13, 24, 0)	1.9%	0.47%	1.4%
2015–2016	34 (12, 21, 1)	1.7%	0.47%	1.2%
2016–2017	32 (12, 20, 0)	1.7%	0.37%	1.3%
2017–2018	31 (12, 17, 2)	1.5%	0.46%	1.0%
2018–2019	25 (7, 18, 0)	1.3%	0.25%	1.1%

TABLE 3. Mechanism of Injury and Player Activity at the Time of Injury for All 208 Injuries

Mechanism or Activity	n (%)
Mechanism of injury	
Player-to-player contact	162 (79%)
Tackling	55 (26%)
Blocking	53 (25%)
Being blocked	29 (14%)
Being tackled	13 (6%)
Other or fighting	12 (6%)
Player contact with ground	46 (21%)
Player position activity at the time of injury	
Defensive back	38 (18%)
Special teams	38 (18%)
Linebacker	29 (14%)
Defensive lineman	27 (13%)
Offensive lineman	25 (12%)
Running back	21 (10%)
Tight end	12 (6%)
Wide receiver	12 (6%)
Quarterback	6 (3%)

TABLE 4. Fracture Characteristics of All 208 Injuries

Characteristic	n (%)
Articular involvement	
Extra-articular	197 (95%)
Intra-articular	11 (5%)
Number of metacarpals fractured per injury occurrence	
Single	197 (95%)
Multiple	7 (3%)
Other*	4 (2%)
Metacarpal injured (isolated injuries, n = 197)	
Thumb	26 (13%)
Second	42 (21%)
Third	50 (25%)
Fourth	62 (31%)
Fifth	17 (9%)
Treatment	
Operative	81 (39%)
Nonsurgical	127 (61%)

*Third metacarpal with ipsilateral third proximal phalanx (n = 1); second metacarpal fracture with ipsilateral second proximal phalanx fracture (n = 1); first metacarpal with thumb ulnar collateral ligament injury (n = 1); and third metacarpal fracture with triangular fibrocartilage complex tear (n = 1).

patients, 12 were managed operatively and the median time to return to play was 24 days (IQR, 16–35 days). This was less than the 55-day (IQR, 12–61 days) return-to-play time for the 6 players with thumb metacarpal fractures who were managed nonsurgically ($P < .05$). A regression analysis revealed there was no trend or difference with respect to age, time in the league, injury setting (practice vs game), mechanism of injury, articular involvement, presence of multiple concomitant injuries, or player position.

DISCUSSION

Metacarpal fractures in American professional football players represented a common and consistent source of player morbidity over the 7 NFL seasons from 2012 through 2018, with a median of 15 days lost from competition after injury. Though surgical stabilization of index- through small-finger

TABLE 5. Missed-Time Injuries: Characteristics, Management, and Return to Play

Characteristic	n (%)	*P* Value
Missed time following injury		
No	43 (21%)	
Yes	157 (75%)	
Unknown	8 (4%)	
Management of players who missed time following injury (n = 157)		
Operative	75 (48%)	
Nonsurgical	82 (52%)	
Management of players who did not miss time following injury (n= 43)		
Operative	4 (9%)	
Nonsurgical	39 (91%)	
Return to play for players who missed time (n = 157)		
Same season	130 (83%)	
Following season	10 (6%)	
Unknown	17 (11%)	
Time of injury for those who missed time but returned the same season (n = 130)		
Preseason	53 (41%)	
Regular season	76 (58%)	
First 8 games	38 (29%)	
Second 8 games	38 (29%)	
Postseason	1 (1%)	
Days to return to play for those who missed time but returned the same season (n = 130)		.2
Total	18 (IQR,10−39)	
Operative (n = 61)	20 (IQR, 16−42)	
Nonsurgical (n = 69)	16 (IQR, 6−30)	

metacarpal fractures did not have an impact on the return-to-play time, the period away from football was significantly extended for players with thumb metacarpal fractures, and even more so for thumb metacarpal fractures treated nonsurgically compared with those treated surgically.

The few published retrospective studies on metacarpal fractures in football players are either very limited in size and scope; profile young, amateur athletes; or lack the detail and specific data from which to draw conclusions, especially around treatment and its impact on return to sport.[4,7,10−12] Rettig et al[2] studied a group of 53 young, amateur athletes from different sports, including 29 football players. Overall, 82% of fractures were minimally displaced or nondisplaced and deemed stable. For football players, there were no cases of multiple metacarpal fractures or concomitant ipsilateral hand injuries. The return-to-play time averaged 10.6 days following nonsurgical care, compared to 13.6 days for the 5 operated football players.[2] Etier et al[11] reviewed the outcomes of 11 in-season, amateur football players, all of whom underwent surgical fixation of metacarpal fractures. Two players sustained 2 metacarpal fractures, but no other ipsilateral hand injuries were reported. The average follow-up was limited to 3 months, and there was no nonsurgical comparison group. These 11 players returned to preinjury participation levels at a mean of 6.3 days, using a protective orthosis for an average of 21 days after surgery.[11]

The only relatively recent study focused on professional football players reviewed epidemiologic data from 1996 through 2005 gathered from the surveillance database in use at that time.[8] Separate hand, first ray, and finger injury cohorts were investigated, but no specific injury episode or individual player−level information was available. There were 241 lesser ray metacarpal fractures recorded, and this was found to be the most

TABLE 6. Breakdown of Players Who Missed Time But Returned to Play in Same Season (n = 130) Based on Metacarpal Injured and Operative Management

Characteristic	n (IQR)	*P* Value
Metacarpal fractured (n = 130)		
Thumb metacarpal, n (%)	18 (14%)	
Nonthumb metacarpal, n (%)	112 (86%)	
Days to return to play based on metacarpal fractured		.03
Thumb metacarpal	29 (16–58)	
Nonthumb metacarpal	17 (8–30)	
Days to return to play based on management for thumb metacarpal fracture (n = 18)		.01
Operative (n = 12)	24 (16–35)	
Nonsurgical (n = 6)	55 (12–61)	
Days to return to play based on management for nonthumb metacarpal fractures (n = 112)		.1
Operative (n = 49)	19 (15–42)	
Nonsurgical (n = 63)	15 (5–29)	
Days to return to play based on nonsurgical management		.02
Thumb metacarpal (n = 6)	55 (12–61)	
Nonthumb metacarpal (n = 63)	15 (5–29)	
Days to return to play based on operative management		.17
Thumb metacarpal (n = 12)	24 (16–35)	
Nonthumb metacarpal (n = 49)	19 (15–42)	

frequently sustained injury within the hand group (80%).[8] As part of the first ray group, 9 extra-articular thumb metacarpal fractures and 34 Bennett's fractures were reported, representing 3% and 12% of all metacarpal fractures, respectively. The overall average return-to-sport times were listed separately for hand (lesser digit), thumb, and Bennett's metacarpal fractures as 16, 14, and 30 days, respectively.[8] An accurate return-to-play assessment required specific injury episode and treatment information that was not available, including knowledge of coexistent injuries and determinations of medians and IQRs. The purpose of this present investigation was to leverage the extensive, league-wide data provided through the NFL's ISS and the EMR to more precisely define the occurrences and variables associated with metacarpal fractures in NFL players and detail the specific mechanisms of injury, while identifying susceptible players and determining the primary factors which impact their return to play.

Commonalities were noted between the present investigation and previously published investigations.[2,7,11] Extra-articular third and fourth metacarpal fractures were the most frequently encountered fractures, with the majority being isolated injuries. Most injuries happened in game settings and were caused by player-to-player contact while blocking or tackling, and injuries frequently involved defensive backs. In contrast to the Mall et al[8] and Etier et al[11] studies, we found metacarpal fractures to be common in special team players, yet infrequent in receivers. Our findings also indicate that the absolute number of metacarpal fractures, as well as the risk of fracture, is highest in the regular season.

For the purpose of characterizing the more severe missed-time injuries and evaluating factors likely to have an impact on the return-to-play time, we defined a cohort of 130 players, representing those NFL athletes who sustained a metacarpal fracture and missed playing time but returned to football the same season. This eliminated players with less severe injuries who did not require time away, as well as players whose injuries ended their seasons, thereby providing no defined return-to-sport time. Only 10 players did not return the same season. Eight of these 10 athletes sustained the fracture in week 12 or later, with 4 injured in week 17 or in the postseason. Six players were treated surgically, but no one in this 10-player subset experienced concomitant ipsilateral hand or wrist injuries or intra-articular fractures. One sustained a thumb fracture. Of the 2 players who were injured prior to week 12 and did not return, 1 underwent surgery. There was no significant difference in return to play for these 130 players when comparing nonsurgical care to surgical treatment,

except in the case of thumb metacarpal fractures, where surgery significantly decreased the return-to-sport time. Despite the frequency of surgery for intra-articular fractures being low, the operative rate was 100% for multiple metacarpal fractures and for isolated fractures with concomitant, ipsilateral hand or wrist injuries; these features are suggestive of more extensive trauma and an extended return-to-play time, irrespective of the treatment choice. Return-to-football treatment comparisons may have been influenced by the need for surgery in these more severe cases. In most cases, both operative and nonsurgical fracture management allowed for return to play less than 3 weeks after the injury. From a fracture location perspective, only involvement of the first ray had an impact on the return-to-play time. Furthermore, players sustaining first ray metacarpal fractures had a significantly higher likelihood of undergoing operative management than those fracturing other rays.

There are several strengths to this study, most notably the use of a database characterized by standard data collection, curation, and reporting on an injury-episode basis across the entire NFL and including all players. An expanded ISS, with a renewed emphasis on reporting, was mandated for all NFL teams prior to the 2012 season. Injury collection still followed a "reportable injury definition" and missed playing time to trigger injury entries, however, all fractures were considered critical, separately reportable injury occurrences, irrespective of the time lost. Injury reporting and surveillance was melded with clinical management before the 2015 season with the introduction of the current EMR system, tailored to orthopedic, neurologic, and internal medicine disorders. It fully chronicles player injuries and, through unique identifiers, links other sport-related data, including extensive game statistics and roster information.[9] In this study, the first 3 seasons (2012−2014) relied on data obtained from the ISS, whereas the final 4 seasons (2015−2018) used data from the expanded NFL EMR. The inherent data variation resulting from this change was mitigated by fractures being considered separately reportable injury episodes even in 2012. In fact, the findings from the first 3 years to the last 4 years were consistent. The number of injuries, percentage of fractures requiring surgery, mechanisms of injury, return-to-play times, and all other primary statistics were constant. However, the percentages of players who missed no time following their injury from 2012 through 2014 versus from 2015 through 2018 were 8% and 30%, respectively. It is likely some injury episodes in which players did not miss time were not recorded in the ISS despite the designation of a fracture as a separately reportable event. This aligns with the suggestion from Dreyer et al[9] that roughly 20% more injuries were recorded per reportable injury category after instituting the EMR.

There are several important limitations to this study. Information on hand dominance, as well as fracture-specific and treatment-specific details, including the method of operative stabilization and use of postoperative playing orthoses, was not available. There were no operative notes, nor radiographs and reports to review. Information from these sources would have helped characterize the injuries further and explain treatment decisions and return-to-play results. Additionally, return-to-play calculations were hampered by missing data in 35 cases. However, this missing information applied equally both over the years and to those treated by either operative or nonsurgical means. Finally, many factors were considered regarding return to play; however, there are certain scenarios where a player may be held out longer (or returned sooner) than medically recommended based not just on fracture specifics and player disability but also the treating physician's beliefs and experiences, as well as additional, nonmedical factors intrinsic within professional sports.

By leveraging the mandated league-wide database, which included detailed player-level injury data linked to relevant sport-specific numbers, this investigation, and its highly reliable data, provide a unique and thorough analysis of metacarpal fractures and their treatments occurring in NFL athletes from the 2012 through 2018 seasons. The result is a precise recording of fracture occurrences, associated injuries, mechanisms, treatments, and outcomes as judged by the return to sport. Awareness of the variables associated with the occurrence of metacarpal fractures in the NFL should inform preventative actions. The risk of injury is highest during games and in the regular season. The most frequently injured bones are the third and fourth metacarpals. The mechanism of injury is player-to-player contact in the majority of cases and is likely due to direct blows with or without rotational or angular forces imparted to the digits. With this insight, protective equipment worn by special teams players and defensive backs, designed to deflect direct blows and control finger rotation and position, especially for the third and fourth digits, could decrease the incidence of these fractures. If fracture severity results in time lost, surgical stabilization will not necessarily accelerate a return to football, except in cases of thumb metacarpal fractures.

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